

Q. Describe the phases of a Basic Instruction Cycle. Explain how the CPU handles interrupts by modifying the standard instruction cycle.

📌 Definition to Remember

The **Instruction Cycle** is the fundamental operational process of a CPU, consisting of three phases: **Fetch, Decode, Execute**. When an external event demands immediate attention, the CPU modifies this cycle by appending an **Interrupt Cycle** to save state and run an Interrupt Service Routine (ISR).

1. Standard Instruction Cycle Phases

1. Fetch Cycle:

- * CPU fetches the instruction from memory using the address in the Program Counter (PC).
- * Instruction is placed in the Instruction Register (IR).
- * PC is incremented to point to the next instruction.

2. Decode Cycle:

- * Control Unit decodes the Opcode in the IR to determine the operation.

3. Execute Cycle:

- * ALU performs the requested operation. Result is stored in a register/memory.
(Loops back to Fetch).

2. Handling Interrupts

An **Interrupt** breaks the normal sequential flow. To handle it, the CPU adds an **Interrupt Cycle** immediately after the Execute cycle.

Interrupt Processing Steps:

1. **Check:** At the end of every Execute cycle, CPU checks if any interrupt flag is set. If no, fetch next instruction.
2. **Suspend & Save State:** If interrupt is pending, CPU suspends the current program and pushes the **Program Counter (PC)** and Status Flags onto the **Stack** (saves context).
3. **Load ISR:** CPU loads the PC with the starting address of the **Interrupt Service Routine (ISR)**.
4. **Execute ISR:** CPU fetches and executes the ISR to handle the event (e.g., reading a key press).
5. **Return:** The ISR ends with **IRET** (Interrupt Return). CPU pops the original PC and Flags from the stack and seamlessly resumes the main program.

★ Must-Write Points (for 10 marks)

1. Standard cycle: **Fetch → Decode → Execute**.
2. **Fetch:** Gets instruction via PC, stores in IR, increments PC.
3. **Decode:** Control unit translates the Opcode.
4. **Execute:** ALU performs the task.
5. **Interrupt Cycle** occurs at the end of the Execute phase.
6. CPU suspends program and saves the **Program Counter (PC) and Flags to the Stack**.

7. CPU executes the **Interrupt Service Routine (ISR)**, then uses **IRET** to pop the PC and resume the original program.

Quick Recall

Cycle: Fetch (PC to IR) → Decode (CU) → Execute (ALU) → Check Interrupt → Save PC to Stack
→ Run ISR → IRET (Pop PC) → Resume